



## Screening of Media Supplements for High-Performance Perfusion Cultures by Design of Experiment

Patrick Mayrhofer and Renate Kunert

### Abstract

Perfusion is considered as the preferable unit operation mode for fully integrated continuous bioprocessing. However, the inherent complex process control, long process development times, and lack of suitable scale-down models for high-throughput screening are reasons why perfusion processes are still not routinely applied in cell culture technology. Advantages of perfusion are maintenance of a consistent cellular environment, a constant high-quality product flow, enhanced volumetric bioreactor productivity, and small lab footprint. Here, we provide guidelines for screening different proprietary but commercially available HyClone™ Cell Boost™ supplements in a Design of Experiment (DoE) approach to spike the HyClone™ CDM4NS0 basal media for enhanced product titers in small-scale TubeSpin models. These surrogate semi-perfusion cultures were successfully realized by a daily complete media exchange routine resulting in high viable cell densities for extended time periods at minimal media consumption. This technique was leveraged to define the potential of different perfusion media formulations.

**Key words** Mammalian cell culture, Perfusion culture, Scale-down model, Small-scale bioreactor, DoE, Response surface methodologies, Process development

---

## 1 Introduction

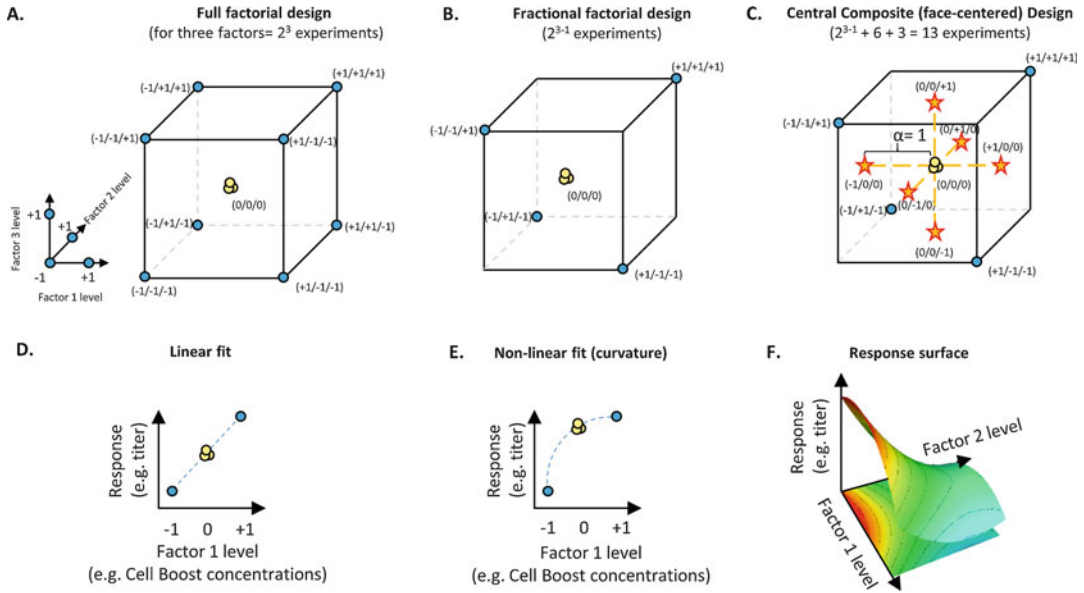
Perfusion bioprocesses in mammalian cell culture technology are characterized by a continuous flow of a nutrient-rich media into and a cell-free spent media flow out of the bioreactor. Thereby, the cells encounter an “open” or “closed” cell retention device [1–3]. This enables a continuous product flow that makes perfusion especially suitable as valuable unit operation integrated in automated and fully continuous biomanufacturing workflows [4–7]. Scaling of total protein production is realized by increasing the fermentation time rather than volumetric expansion, which allows to keep bioreactor equipment and lab footprint small [8]. As a consequence, the same established process can be used for producing variable amounts of therapeutic proteins to address the demand at different clinical stages. Because of the small

bioreactor size, such processes can be performed in customized single-use equipment. This highly flexible operation mode keeps the cells in a healthy, physiological state by continuously removing spent media containing toxic by-products. The harvested flow is counterbalanced by a rich, but balanced, perfusion medium for continuous replenishment of required nutrients.

Culture media optimized for perfusion processes are characterized by achieving high cell densities and productivities at minimal media consumption rates [9, 10]. Such media quality attributes may be achieved by a balanced nutrient mix supporting critical media properties such as osmolality, pH, and buffer capacity. The objective is to obtain minimal cell-specific perfusion rates (CSPR, expressed as picoliter per cell per day) supporting high cell viabilities and specific productivities [4]. CSPR can be used as a benchmark value to compare the performance of different perfusion media. Approaches for development of perfusion media include blending approaches to spike basal media with different media supplements in combination with spent media analyses to identify and to overcome specific nutrient limitations. Thereby, already optimized fed-batch feed solutions can be repurposed for use in perfusion processes [10].

Due to the operational complexity and long-process runs, perfusion process development typically is very time-consuming. Another issue hampering fast perfusion medium and process development is the lack of suitable small-scale models that can be used under high-throughput conditions. Successful attempts for perfusion culture optimization were performed in the past by using manual models in daily discontinuous batch-refeed operation [11–13]. Automated parallel multivessel systems were developed and are commercially distributed. The first of such small-scale parallel bioreactor systems applied in simulated perfusion modes used sedimentation as cell separation strategy [14]. Currently, an AMBR250 system for working volumes up to 250 mL for 24 parallel minibioreactors in real perfusion mode is also available. However, such equipment usually requires high investment costs and therefore is not always available for the broader scientific community.

Media development usually requires multiple parallel screening of many different conditions. Therefore, the experimental design is crucial to reduce the number of experiments and operators, but to extract maximum information. Design of Experiment (DoE) and response surface methodologies (RSM) are suitable methods to identify, model, and optimize the impact of different “factors” that influence so-called responses. Different media supplements [15] can be chosen to represent such factors at different concentrations (“factor levels”). The measured responses of interest can be defined as distinct values for product titer or total cell concentration. The advantage of such DoE strategies as opposed to



**Fig. 1** Combination of different experimental designs to establish the central composite (face-centered) design for calculation of the final response surface model. A full factorial design for three factors at two levels ( $2^3$  experiments) is plotted as a cube (a). A (half) fractional factorial design ( $2^{3-1}$  experiments) is used to reduce the number of experimental runs (b). For modeling the main-, interaction-, and quadratic effects, a central composite design is applied by supplementing the fractional factorial design with center points (yellow) and axial star points at a star distance of  $\alpha = 1$  in a face-centered design (c). Center points (yellow) are highly suitable to detect a linear (d) or nonlinear factor-response relationship (e). The latter is represented by a response surface for two or more factors (f)

One-Factor-At-A-Time approaches (“OFAT”) lies in the detection of factor interactions, reduced experimental expenditure, and statistical confidence.

Modeling of the impact that different factor levels have on the responses is mandatory, because the true relationship is unknown and in reality only a subset of chosen factor levels and measured responses can be performed. In a full factorial design, all factor combinations are compared to each other to find main factor effects and factor interactions (Fig. 1a). A fractional design can be used to reduce the experimental investment (Fig. 1b). In a typical DoE experiment, the first step is to decide about a first rough “screening” experimental design or a more-detailed “response surface” design. The “screening” experimental design identifies the few main factors and their interactions on a given response. The “response surface” design characterizes and models the relationship between the chosen quantitative factors and the measured responses by regression analysis in a central composite design (Fig. 1c). The latter option represents an “optimization” experiment to find the optimum, i.e., an optimal combination of different factor levels for maximized responses in the chosen design space.

Usually, the impact of one or several factors is not described by a simple linear relationship (Fig. 1d). Moreover, a measured response on certain factor levels is most often described by a curve for one factor (Fig. 1e) or by a “response surface” for two or more factors (Fig. 1f). This means that in such experiments, curvature in the response surface must be detected. This is accomplished by using center points (compare yellow center points in Fig. 1d, e) and to finally model the response surface correctly with a second-order polynomial (“quadratic”) function. This enables to calculate (1) the linear (main) effects, (2) factor interactions, and (3) quadratic effects and finally to define the optimal factor combinations of the investigated experimental space.

Here, we demonstrate the use of DoE for spiking different commercially available HyClone™ Cell Boost™ supplements into the HyClone™ CDM4NS0 basal media for optimal semi-perfusion processes in small-scale shaking tubes. This workflow is highly suitable to empirically optimize media compositions when the experimental operator does not know the proprietary ingredients.

---

## 2 Materials

1. Class 2 biological safety cabinet.
2. Serological pipette.
3. Cell line of interest: recombinant CHO-K1 cell line expressing an IgG1 antibody.
4. Basal medium: HyClone™ CDM4NS0.
5. HyClone™ Cell Boost™ supplement 1, 3, 7a, and 7b.
6. TubeSpin minibioreactors.
7. Kuhner™ shaker incubator (37 °C, 220 rpm, 50 mm orbital shaking diameter, 7% CO<sub>2</sub>, 85% relative humidity).
8. Analytical instruments: ViCell cell viability analyzer, ELISA, Octet, Biacore or HPLC for titer determination, Bioprofiler or Cedex for metabolites (optional), and osmometer (optional).
9. MODDE software, version 12.1.

---

## 3 Methods

In this example, the impact of HyClone™ Cell Boost™ supplements, chosen as the investigated DoE factors, on the steady-state mean titer and mean total cell concentration responses was investigated. Four high-performing HyClone™ Cell Boost™ supplements (CB1, 3, 7a, and 7b) were already preselected in a first

DoE screening design by reducing a full factorial design (Fig. 1a) to a fractional factorial design at two levels (Fig. 1b) (data not shown) (*see Note 1*). Here, we present the pre-experimental planning to define the factor levels and the design space, the actual cell culture handling, and the post-experimental regression analysis to model the responses and to find the optimal combination of the four HyClone™ Cell Boost™ supplements using a central composite face-centered design (Fig. 1c).

### 3.1 Design of Experiment (DoE)

We defined the optimal “factor level” combination for a maximized titer response as the primary experimental design objective. It is reasonable to assume that there is curvature in the factor-response relationship, and therefore, a suitable experimental design to establish a response surface was chosen. Therefore, a fractional factorial design combined with center points and axial “star points” was used in a central composite design to detect and estimate such curvature in the response surface. The final experimental design for the four different factors (i.e., Cell Boosts) consisted of a total of 23 runs comprising three replicate center points, eight star points, and twelve runs for the fractional factorial design (Table 1).

1. Start a new experimental design using the MODDE software.
2. Define factors by choosing factor names, abbreviations, units, and type (*see Note 2*). Click next and define the responses (*see Note 3*).
3. Click next and select “Optimization (RSM)” as the chosen objective (*see Note 4*).
4. Click next and choose the “Reduced CCF (star distance: 1)” as the second recommendation.

Curvature can be detected by using three factor levels ( $-1$ ,  $0$ , and  $+1$ ). However, since a  $3^4$  full factorial design with a total of 81 culture runs would be impracticable, a central composite design can be used to reduce the number of experiments to only 23 individual runs when higher-order interactions can be assumed negligible (Table 1). The chosen reduced central composite face-centered design (CFD) consists of (1) twelve runs of a fractional factorial design at two levels, extended by (2) three runs for a center point as replicates (triplicates) and (3) eight axial (star) points with a star distance of  $\alpha = 1$  (*see Note 5*).

### 3.2 Definition of DoE Levels $-1$ , $0$ , and $+1$

DoE levels for each factor should be carefully chosen not to negatively influence the final output (i.e., responses). In this experiment, the total amino acid concentration and osmolality were chosen as criteria to define the three different DoE levels  $-1$ ,  $0$ , and  $+1$ . All Cell Boosts™ were prepared as stock solutions from powder formulations as suggested by the manufacturer. Therefore, Cell

**Table 1**  
**Overview of the used DoE matrix for a central composite (face-centered) design**

| Exp. no.             | CB1 | CB3 | CB7a | CB7b | Design                                         |
|----------------------|-----|-----|------|------|------------------------------------------------|
| 1 (no CB)            | -1  | -1  | -1   | -1   | Cube points of the fractional factorial design |
| 2                    | +1  | -1  | -1   | -1   |                                                |
| 3                    | -1  | +1  | -1   | -1   |                                                |
| 4                    | +1  | -1  | +1   | -1   |                                                |
| 5                    | -1  | +1  | +1   | -1   |                                                |
| 6                    | +1  | +1  | +1   | -1   |                                                |
| 7                    | -1  | -1  | -1   | +1   |                                                |
| 8                    | -1  | +1  | -1   | +1   |                                                |
| 9                    | +1  | +1  | -1   | +1   |                                                |
| 10                   | -1  | -1  | +1   | +1   |                                                |
| 11                   | +1  | -1  | +1   | +1   |                                                |
| 12 (all max. CB)     | +1  | +1  | +1   | +1   |                                                |
| 13                   | -1  | 0   | 0    | 0    | Axial (star) points                            |
| 14                   | +1  | 0   | 0    | 0    |                                                |
| 15                   | 0   | -1  | 0    | 0    |                                                |
| 16                   | 0   | +1  | 0    | 0    |                                                |
| 17                   | 0   | 0   | -1   | 0    |                                                |
| 18                   | 0   | 0   | +1   | 0    |                                                |
| 19                   | 0   | 0   | 0    | -1   |                                                |
| 20                   | 0   | 0   | 0    | +1   |                                                |
| 21–23 (half-max. CB) | 0   | 0   | 0    | 0    | Center points (triplicate)                     |
|                      | 0   | 0   | 0    | 0    |                                                |
|                      | 0   | 0   | 0    | 0    |                                                |

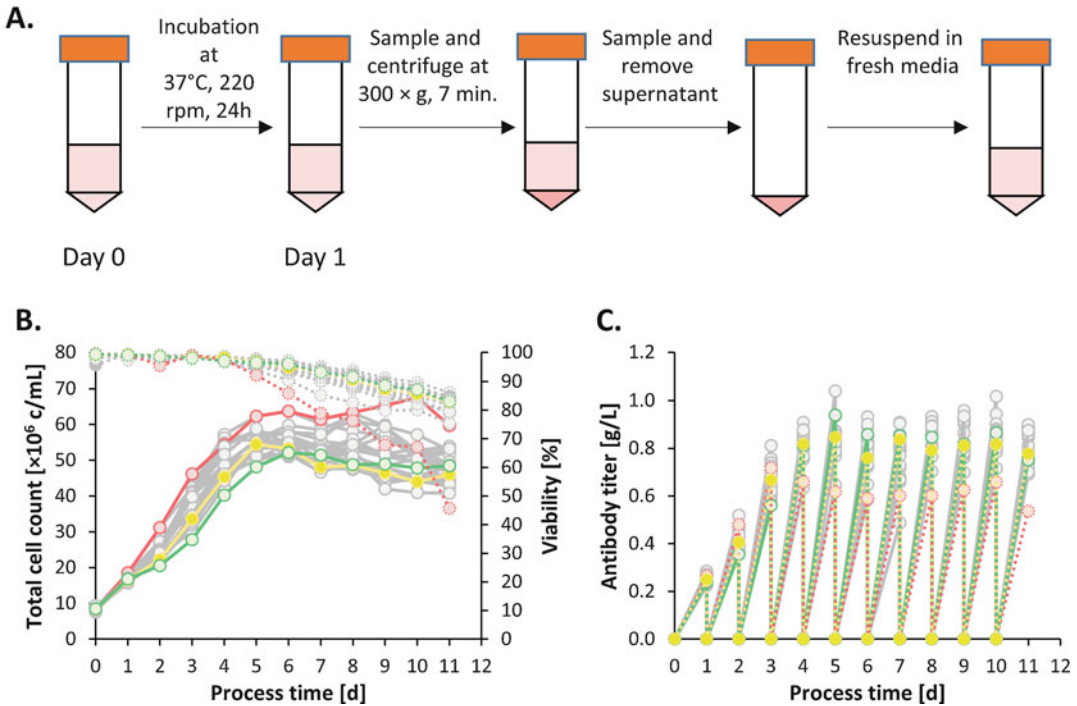
The final design consists of a total of 23 experimental runs, including 12 runs for the fractional factorial design at 2 levels, 8 for the axial star points and triplicate cultures for the center point

Boost™ 1, 3, 7a, and 7b were prepared as 10%, 5%, 18.1%, and 9.5% stock solutions, respectively. In the next step, the molarity of total amino acids of the respective stock solutions was determined. The four individual Cell Boosts™ were then mixed and normalized to their total molar amino acid concentration (*see Note 6*), meaning that the molar contribution of amino acids from each Cell Boost™ is the same. The CDM4NS0 basal media was then spiked with this Cell Boost™ mix to reach a maximum osmolality of 400 mOsm/kg. This critical osmolality threshold was chosen for this specific cell line and should be empirically determined for other cell lines. The maximum spike concentration to reach 400 mOsm/kg was defined as level +1 for each individual Cell Boost™. Level -1 was defined as no Cell Boost™ addition, and level 0 was the half-maximum spike addition. Importantly, this normalization strategy is applicable to any other proprietary basal media/feed solution system on the market, provided that the total amino acid concentration of the supplements is known or analyzed by straightforward standard HPLC protocols. Accordingly, for all Cell Boost™ solutions at level +1, a 10 mL aliquot of basal medium was spiked with 1.106 mL Cell Boost™ 1, 1.99 mL Cell Boost™ 3, 0.381 mL Cell Boost™ 7a, and 0.076 mL Cell Boost™ 7b (*see Note 6*).

### 3.3 Cell Culture Experiments (“Semi-Perfusion” Cultures)

Small-scale shaking tube cultures are suitable cultivation systems that can be handled by a single operator for more than 20 different conditions simultaneously. A daily, complete media exchange was performed to simulate a perfusion process in the culture tubes and to supply cells with fresh media, but to remove spent media containing inhibiting toxic by-products such as lactate and ammonia (Fig. 2a). Because very high cell densities above  $50 \times 10^6$  c/mL can be reached using cell retention by centrifugation (Fig. 2b), it is important that a strict 24 h sampling and media exchange interval is followed and planned accordingly. It is further recommended that sampling time is kept rather short so that cells do not settle, to ensure homogeneous suspensions and representative cell numbers. Thus, it is recommended that only four tubes are sampled in parallel before returning them back into the shaker incubator. Cultures should also be processed rather fast after centrifugation because it was shown that local pH and dissolved oxygen (DO) critically decrease in short time, especially when high cell numbers are present as a cell pellet. Using this method, more than  $50 \times 10^6$  c/mL were reached after five days and supernatant with antibody titers of more than 0.8 g/L could be harvested daily, after each media exchange (Fig. 2c).

1. Prepare an inoculum culture to reach  $10 \times 10^6$  c/mL in finally 10 mL for each condition tested. For a total of 23 tubes, thus, seed two 1 L Erlenmeyer shake flasks with 300 mL at  $0.5 \times 10^6$  c/mL.



**Fig. 2** Semi-perfusion runs in TubeSpin mini-bioreactors. Cultures according to a DoE matrix (Table 1) were seeded at  $10 \times 10^6$  c/mL, and a daily sampling, centrifugation, and media exchange protocol was followed (a). Using this method, high cell densities of more than  $50 \times 10^6$  c/mL at high viabilities could be reached within 5 days (b), producing daily antibody titers of more than 0.8 g/L (c)

With an assumed specific growth rate of  $\mu = 0.7 \text{ day}^{-1}$  (approx. one doubling in 24 h) and a 3-day culture period (from Friday to Monday), this will accumulate to  $0.5 \times 10^6 \text{ c/mL} \times 600 \text{ mL} \times e^{(0.7 \text{ day}^{-1} \times 3 \text{ days})} = 2450 \times 10^6 \text{ cells}$ , enough to seed 23 tubes at  $10 \times 10^6 \text{ c/mL}$  in 10 mL (total of  $23 \text{ tubes} \times 10 \text{ mL/tube} \times 10 \times 10^6 \text{ c/mL} = 2300 \times 10^6 \text{ cells}$ ). The final volume of the required expansion culture depends on the actual growth rate of the used cell line.

2. Prepare the differently spiked media according to the DoE matrix in Table 1 (see Note 7).
3. Aliquot a total of  $100 \times 10^6$  cells into each semi-perfusion tube, centrifuge at  $300 \times g$  for 7 min, and resuspend cells in 10 mL fresh and pre-warmed CDM4NS0 basal medium spiked with the respective Cell Boosts™ to start the semi-perfusion process (see Note 8).
4. Tubes are cultured at 220 rpm in a Kuhner shaker incubator at 50 mm orbital diameter.



5. Start the daily sampling routine by transferring 100  $\mu\text{L}$  of cell suspension into a ViCell sample tube and dilute with 600  $\mu\text{L}$  PBS (dilution factor: 7) for determination of cell number and viability using the ViCell cell viability analyzer (*see Note 9*).
6. Centrifuge the cultures at  $300 \times g$  for 7 min, and sample culture supernatant aliquots for determination of antibody titer and metabolite concentrations. Centrifugation for medium exchange is performed in the same culture tube unless excessive accumulation of cell debris at the side of the vessel wall is observed. Otherwise, the vessel is changed if needed in order to avoid accumulation of dead cells or debris derived from precipitates at the vessel walls.
7. Remove the remaining supernatant by decanting (*see Note 10*).
8. Resuspend the cell pellet in 10 mL of fresh, pre-warmed, media according to Table 1 (*see Note 11*).
9. Repeat **steps 5–8** for subsequent culture days.

### 3.4 Regression Analysis by Response Surface Methodologies (RSM)

The combination of a (fractional) factorial design together with center and star points is called “central composite design.” By extending the fractional factorial design with axial (“star”) points and replicate center points (Fig. 1c), quadratic effects can be estimated by building a second-order (quadratic) polynomial regression model. For a response surface including 4 factors, a quadratic (2nd-order) polynomial function in order to get 15 model coefficients ( $\beta_0$  to  $\beta_{14}$ ) explaining the intercept ( $\beta_0$ ), main factor effects ( $\beta_1$  to  $\beta_4$ ), factor interactions ( $\beta_5$  to  $\beta_{10}$ ), and quadratic effects ( $\beta_{11}$  to  $\beta_{14}$ ):

$$\begin{aligned}
 y = & \beta_0 + \beta_1 \times x_{CB1} + \beta_2 \times x_{CB3} + \beta_3 \times x_{CB7a} + \beta_4 \times x_{CB7b} \\
 & + \beta_5 \times x_{CB1} \times x_{CB3} + \beta_6 \times x_{CB1} \times x_{CB7a} + \beta_7 \times x_{CB1} \\
 & \times x_{CB7b} + \beta_8 \times x_{CB3} \times x_{CB7a} + \beta_9 \times x_{CB3} \times x_{CB7b} + \beta_{10} \\
 & \times x_{CB7a} \times x_{CB7b} + \beta_{11} \times x_{CB1}^2 + \beta_{12} \times x_{CB3}^2 + \beta_{13} \\
 & \times x_{CB7a}^2 + \beta_{14} \times x_{CB7b}^2 + \varepsilon
 \end{aligned} \tag{1}$$

$y \rightarrow$  responses (antibody titer or total cell count)

$\beta \rightarrow$  model coefficients

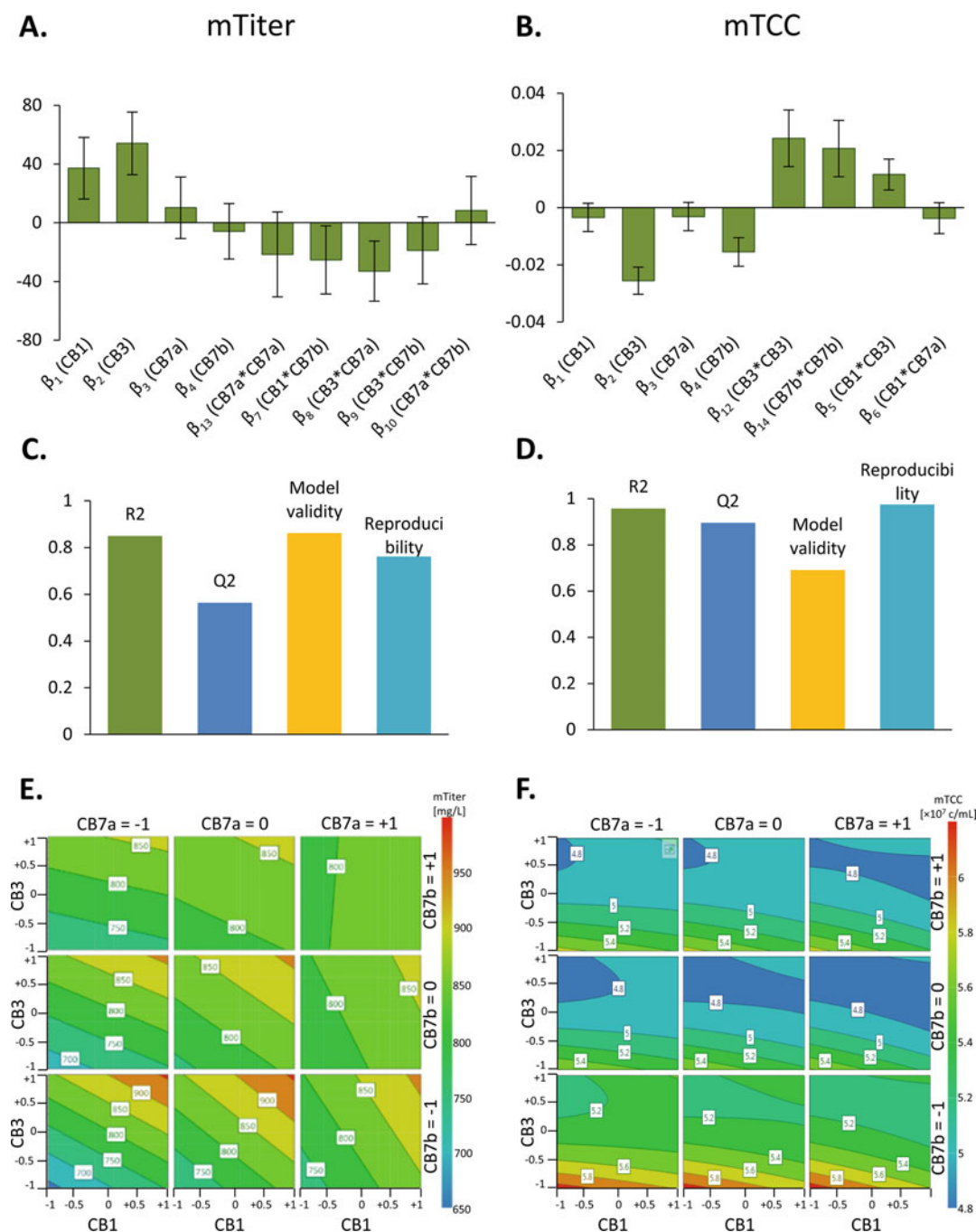
$x \rightarrow$  factor levels  $-1$ ,  $0$ , or  $+1$  (i.e., Cell Boost™ concentrations)

$\varepsilon \rightarrow$  error

The coefficients are optimally varied by the software MODDE so that the fit of the model is optimized for the experimental data points. Detailed analyses and model generation can be performed for multiple critical cell culture attributes (responses) considering peak or average values across multiple days. Peak values are prone for outliers and should be carefully evaluated. Using this semi-

perfusion approach, stable steady-state values are obtained. It is recommended to calculate average values once such a steady state is reached for all cultures (day 4 to 11).

1. Manually pre-evaluate the obtained raw data for obvious outliers or unusual behavior.
2. Calculate the critical cell culture data such as peak titer (pTiter), mean titer (mTiter), cumulative titer (cTiter), viable cumulative cell days (VCCD), peak total cell count (pTCC), mean total cell count (mTCC), maximum specific productivity (qP), and mean specific productivity (mqP).
3. Start the modeling in MODDE by clicking the “Analysis Wizard.”
4. A replicate plot is presented by the program showing the obtained responses for each culture. The overall variability of the different tested conditions should be much greater than the variability within the triplicate center points. Go to the next window by clicking “Next>.”
5. The shape of the response distribution is highlighted to determine if data transformation is needed when the distribution is not a “bell-shaped” normal distribution.
6. In the next window, the model coefficients are displayed (Fig. 3a, b) as scaled and centered values. Depending on the chosen DoE design (“central composite face design”) and goal (“optimization”), the linear, interaction, and quadratic (“squares”) terms are displayed. A positive value indicates a positive influence on the respective response. The error bars indicate whether the respective model coefficient is statistically significant or not. Error bars in nonsignificant model coefficients cross the zero line. Small and nonsignificant model coefficients can be removed to keep the model complexity low. This is especially important when models from many different factors are built (*see* **Note 12**).
7. Click “Next>” and check the summary of fit graph to get a first impression for the quality of the established model (Fig. 3c, d). R<sup>2</sup> describes the model fit and should be above 0.5. Q<sup>2</sup> is the most critical and most sensitive indicator for a good model and is an estimate of future prediction precisions. It should be greater than 0.5 and better than  $0.7 \times R^2$  for a good model. Model validity indicates the result of different statistical tests performed by the software to detect outliers, an incorrect model, or transformation problems. This value might also be small if the variability of the triplicates is very small, but still a valid model is obtained. The reproducibility describes the variability of the replicate cultures in comparison to total variability and should be above 0.5.



**Fig. 3** Model output provided by the MODDE software. Using a central composite design, the quadratic model coefficients for mean titer (mTiter) (a) and mean total cell concentration (mTCC) (b) can be calculated. Those values are “center and scaled” by dividing the coefficients by the standard deviation of their respective response. Thereby, coefficients between responses of different ranges are comparable. Model quality for mTiter (c) and mTCC (d) is presented by R2 (green), Q2 (dark blue), the model validity (yellow), and reproducibility (light blue). Optimal factor level combinations for maximum predicted mTiter (e) and mTCC (f) can be illustrated by using contour plots

8. Click “Next>” to obtain the normal probability plot that can be used for the detection of outliers.
9. By clicking “Next>,” the graph for observed vs. predicted values is presented.
10. Repeat **steps 4–9** for generating further models for additional investigated responses.
11. Determine the optimum level factors by using contour plots (Fig. 3e, f) or using the MODDE optimizer function that calculates the optimal factor levels for two or more simultaneously optimized responses.

---

## 4 Notes

1. To detect curvature of the response function, a triplicate center point can be used at this step already.
2. Here, we can freely control the different levels of the quantitative concentration factors using  $-1$  as the low and  $+1$  as the high limits.
3. Further responses can be added later.
4. RSM stands for “response surface methodologies.”
5. In this special case,  $\alpha = 1$  was chosen comprising only three different factor levels ( $-1$ ,  $0$ , and  $+1$ ) in the final design to reduce the media preparation time and complexity. Choosing this design allows a reduction in the number of experiments to a total of 23 runs as opposed to  $3^4 (=81)$  for a full factorial design with four factors at three levels. A star distance  $\alpha = 1$  is called a “face-centered design.” The advantage is that only three factors have to be investigated.
6. Using the total molar amino acid ratio is one option to normalize different feed supplements relative to each other as exemplified in this protocol. However, this requires the knowledge of the amino acid content of the (proprietary) feed supplements. The amino acid content of biological solutions is routinely analyzed by HPLC analysis. In our example, Cell Boost 1 has an amino acid molarity 1.8 times higher than Cell Boost 3, and therefore, we used Cell Boost 3 with a 1.8-fold volumetric excess.
7. Phenol red might be added as pH indicator. Additionally, L-glutamine (4–8 mM) supplementation is needed for certain cell lines.
8. Alternatively, cells can be seeded at  $0.5 \times 10^6$  c/mL directly in the culture tubes for expansion, and semi-perfusion is then started on day 3. This saves additional culture vessels for cell expansion. However, using this method, there might be

increased variability between cultures because of slightly different cell numbers at the start of perfusion. This variability might be minimized by pooling cells and subsequent aliquoting before starting the semi-perfusion routine.

9. The preset ViCell instrument limit is set to  $10 \times 10^6$  c/mL. Dilute if cell numbers above that limit are expected.
10. Additional supernatant can be saved for further analysis or protein A affinity chromatography.
11. Make sure that cells are fully resuspended and no cell clumps are present. For this, some cell lines can be vortexed shortly. Alternatively, homogeneous mixing can be achieved by pipetting the suspension up and down.
12. In the MODDE software, the “auto-tuning” function helps to optimize the model quality.

## References

1. Woodside SM, Bowen BD, Piret JM (1998) Mammalian cell retention devices for stirred perfusion bioreactors. *Cytotechnology* 28:163–175
2. Voisard D, Meuwly F, Ruffieux P-A et al (2003) Potential of cell retention techniques for large-scale high-density perfusion culture of suspended mammalian cells. *Biotechnol Bioeng* 82:751–765
3. Castilho LR, Medronho RA (2002) Cell retention devices for suspended-cell perfusion cultures. *Adv Biochem Eng Biotechnol* 74:129–169
4. Bielser J-M, Wolf M, Souquet J et al (2018) Perfusion mammalian cell culture for recombinant protein manufacturing – a critical review. *Biotechnol Adv* 36:1328–1340
5. Godawat R, Konstantinov K, Rohani M et al (2015) End-to-end integrated fully continuous production of recombinant monoclonal antibodies. *J Biotechnol* 213:13–19
6. Warikoo V, Godawat R, Brower K et al (2012) Integrated continuous production of recombinant therapeutic proteins. *Biotechnol Bioeng* 109:3018–3029
7. Rathore AS, Agarwal H, Sharma AK et al (2015) Continuous processing for production of biopharmaceuticals. *Prep Biochem Biotechnol* 45:836–849
8. Chen C, Wong HE, Goudar CT (2018) Upstream process intensification and continuous manufacturing. *Curr Op Chem Eng* 22:191–198
9. Ritacco FV, Wu Y, Khetan A (2018) Cell culture media for recombinant protein expression in Chinese hamster ovary (CHO) cells: history, key components, and optimization strategies. *Biotechnol Prog* 34:1407–1426
10. Lin H, Leighty RW, Godfrey S et al (2017) Principles and approach to developing mammalian cell culture media for high cell density perfusion process leveraging established fed-batch media. *Biotechnol Prog* 33:891–901
11. Gomez N, Ambhaikar M, Zhang L et al (2016) Analysis of Tubespins as a suitable scale-down model of bioreactors for high cell density CHO cell culture. *Biotechnol Progr* 33:490–499
12. Villiger-Oberbek A, Yang Y, Zhou W et al (2015) Development and application of a high-throughput platform for perfusion-based cell culture processes. *J Biotechnol* 212:21–29
13. Wolf MKF, Lorenz V, Karst DJ et al (2018) Development of a shake tube-based scale-down model for perfusion cultures. *Biotechnol Bioeng* 115:2703–2713
14. Goletz S, Stahn R, Kreye S (2016) Small scale cultivation method for suspension cells. <https://patents.google.com/patent/WO2016193083A1/en>
15. Parampalli A, Eskridge K, Smith L et al (2007) Development of serum-free media in CHO-DG44 cells using a central composite statistical design. *Cytotechnology* 54:57–68